

Exit Ticket Questions for Week 1

1. Why does $h[n]$ completely characterize an LTI system?
2. Why can an LTI system change the amplitude and phase of a sinusoid, but not its frequency?

Good answers:

1. Any discrete-time signal can be written as a linear combination of $x[k]$ times the basis function $\delta[n - k]$. By linearity, the response to a scaled impulse is scaled and by time-invariance the response to $\delta[n - k]$ is $h[n - k]$. Therefore, again by linearity, $y[n] = \sum_k x[k]h[n - k] = x[n] * h[n]$.

Look for both linearity and time-invariance in the answer.

2. Complex exponentials are eigenfunctions of LTI systems: Input $x[n] = e^{j\omega_0 n}$ produces output $y[n] = H(e^{j\omega_0})e^{j\omega_0 n}$ (due to convolution which depends on linearity and time-invariance). The multiplier $H(e^{j\omega_0})$ can change magnitude and phase, but the factor $e^{j\omega_0 n}$ remains at the same frequency.

Look for eigenfunctions, eigenvalues and convolution (which requires linearity and time-invariance) in the answer.

Answers:

Mediocre answer: Because an LTI system is defined by its impulse response (circular response)

Good answer: Because every input can be decomposed into scaled and shifted impulses, and linearity and time-invariance tell us that the output is the corresponding sum of scaled and shifted copies of $h[n]$.

2. Complex exponentials are eigenfunctions of LTI systems: $x[n] = e^{j\omega_0 n}$ produces $y[n] = \sum_k h[k] e^{j\omega_0(n-k)} = e^{j\omega_0 n} \sum_k h[k] e^{-j\omega_0 k} = e^{j\omega_0 n} H(e^{j\omega_0})$. The eigenvalue $H(e^{j\omega_0})$ is a constant with magnitude and phase and only the factor $e^{j\omega_0 n}$, which is the same for the input and the output of the LTI system (eigenfunction), determines frequency.

Weak answer: Because LTI systems preserve frequency. (true but circular.)

Good answer: Due to the convolution equation $y[n] = \sum_k h[k] x[n-k]$, the output $y[n]$ of an LTI system in response to a rotating phasor $x[n] = e^{j\omega_0 n}$ is a linear combination of scaled and time shifted rotating phasors $h[k] e^{j\omega_0(k-n)}$ of the same frequency as the input phasor.

The core reason: sinusoids are eigenfunctions of LTI systems.

Linearity is what lets you factor $e^{j\omega n}$ out and lets superposition apply — the output at frequency ω depends only on the input at frequency ω , not on cross-terms mixing different frequencies together.

Time-invariance is what makes $H(e^{j\omega})$ a constant instead of a function of n . If the system's parameters changed with time, h would depend on both n and k , and the $e^{j\omega n}$ term wouldn't factor cleanly out of the integral

A good exit ticket answer should be short, correct, to the point and demonstrate a well-reasoned understanding of the question being asked. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., $\delta[n]$ for a discrete-time impulse.